*Quality & Quantity* **35:** 325–341, 2001.
© 2001 *Kluwer Academic Publishers. Printed in the Netherlands.*


# Measuring Agreement in Ordered Rating Scales

CEES VAN DER EIJK
*University of Amsterdam, Department of Political Science, and Amsterdam School of Communication Research (ASCoR); e-mail: vandereijk@pscw.uva.nl*

**Abstract.** Ordered rating scales are one of the most frequently used question formats in large-scale surveys. Analysts of the responses to such questions often find themselves in need of describing the degree of agreement (concentration, consensus) of the answers to such questions. For that purpose they commonly use standard deviations of the response distributions, or measures based on these (such as the coefficient of consensus defined by Granberg and Holmberg, 1988), or the coefficient of variability, etc. This paper demonstrates that such measures are inappropriate for this purpose because they misrepresent what they are supposed to measure: the 'peakedness' of a distribution. As an alternative a measure of agreement A is proposed. This measure is a weighted average of the degree of agreement that exists in the simple component parts – layers – into which any frequency distribution can be disaggregated, and for which agreement can be expressed in a straightforward and unequivocal way.

**Key words:** descriptive statistics, measures of concentration, measures of dispersion, survey research

## 1. The Problem

The degree to which a group of people shares the same preferences or the same perceptions is one of the classic concerns in studies of public opinion, values and political culture, and in questions pertaining to political representation (e.g., McClosky et al., 1960; McClosky, 1964; Pomper, 1972; Converse and Pierce, 1986; Granberg and Holmberg, 1988; Van der Eijk and Franklin, 1996; Van der Brug, 1997; Van der Brug and Van der Eijk, 1999; Thomassen and Schmitt, 1999). How such agreement is most adequately quantified depends on the measurement characteristics of the data involved. Sometimes one can suffice with the percentage of cases on one side of a dichotomy (i.e., agree vs. disagree, left vs. right, post-materialist vs. other, etc.). In other instances, however, such a simple display does not suffice. The extent to which preferences or perceptions are polarized, for example, can not easily be expressed in such a dichotomy, and neither can situations in which high agreement exists, but not with extreme or polar alternatives, but with intermediate ones instead. The specific case addressed in this chapter involves the description of the degree of agreement amongst a group of individuals who express their preferences or perceptions in terms of a number of ordered categories. Ordered rating scales in survey research constitute a characteristic situation: respondents are asked to indicate where they place something in a limited number

of ordered categories. Such rating scales are sometimes used for registering opinions or preferences, as in the case where respondents are asked to indicate which position on a scale defined by two polar opposites best describes their own opinion, viewpoint or preference. To the extent that respondents express the same preference one may speak of *preferential agreement*. In the literature the term consensus is often used when such agreement is high. Ordered rating scales may also be used for registering perceptions. Respondents may be asked to indicate, for example, which position on a left/right scale, on a pro/anti abortion scale or on any similar kind of continuum best characterizes a particular political party or politician. The term *perceptual agreement* refers to the extent to which different people have the same perceptions.

The characteristic that is to be measured by the scale in question may be rather abstract in nature, for example when it is defined by such polar opposites as 'left' and 'right' in many European countries, or 'liberal' and 'conservative' in the American context. Alternatively, it may be more concrete, for example when it refers to specific policy alternatives such as, e.g., prohibiting abortion in all circumstances on the one hand, or allowing it without restrictions on the other.

Typically, ordered rating scales have a relatively small number of categories: 7, 9, 10, and 11-point scales are popular in survey studies when addressing policy or ideological continuums. Scales with 3, 4 or 5 categories are less common, yet far from absent. The coefficient of agreement proposed below is suitable for all these kinds of scales. For a number of reasons that will become obvious below, however, it is less amenable for ordered rating scales that comprise very large numbers of categories, as e.g., sympathy or thermometer scales that consist of 100 points.[1]

The problem of quantification of the degree of agreement between the responses to an ordered rating scale can simply be illustrated by a real-world example. In the Dutch Parliamentary Election Study of 1981, a sample of voters was asked to indicate on a 7-point rating scale which of the positions best characterized the PvdA (Social Democratic Party) and D66 (Democrats–Left Liberals) on a continuum running from 'income differences should not become smaller' (position 1) to 'income differences should become smaller' (position 7). The distribution of the responses is presented in Table I, together with customarily used measures for characterizing a distribution, such as the mean and the standard deviation (van der Eijk et al., 1981: 159–160).

Were one to take the standard deviation as a relevant measure of dispersion (i.e., as the inverse of agreement), Table I would compel to conclude that Dutch voters agree more about the position of D66 than about the position of the PvdA. Yet, inspection of the frequency distributions in Table I could also result in the reverse conclusion. Would the (relative) frequency in the modal category be used as a crude approximation of agreement, Table I shows that the PvdA mode contains more cases (41.6% of the responses) than the mode of D66 (29.3%). The mode, of course, may be considered too crude because it does not reflect anything of the remainder of the distribution.[2] This can be taken into account, however, however,

*Table I.* Relative frequency distributions (percent) of perceptions of positions of PvdA and D66 on a 7-point rating scale (Dutch National Election Study, 1981, $n = 1812$)

| | Distribution of valid responses (%) | |
|---|---|---|
| Position | PvdA | D66 |
| 1 | 2.4% | 1.6% |
| 2 | 2.8% | 2.6% |
| 3 | 3.2% | 8.2% |
| 4 | 6.2% | 21.0% |
| 5 | 13.5% | 29.3% |
| 6 | 30.4% | 27.0% |
| 7 | 41.6% | 10.3% |
| Total | 100% | 100% |
| ($n$) | (1520) | (1318) |
| Mean | 5.83 | 4.96 |
| Stand. dev. | 1.45 | 1.31 |

by considering a particular kind of cumulative frequency distribution. Starting with the modal category one cumulates in successive steps the adjacent category that contains most cases. In other words, one then considers the combination of contiguous categories that combined holds more cases than any other, and increases the number of contiguous categories successively until all categories are included (and hence, the cumulative frequency is 100%). Table II reports for both distributions from Table I the order in which the categories of the 7-point scales are combined, and the cumulated frequencies of each combination.

Table II shows not only that the PvdA mode holds more responses than the one of D66, but also that the most populous combinations of 2, of 3 and of 4 contiguous categories contain for the PvdA considerably more cases than for D66. Only combinations of 5 or 6 adjacent categories contain marginally more cases for D66 than for the PvdA. Overall the pattern implies that agreement about the location of the PvdA is higher than for D66, whereas the standard deviations suggests the opposite. It does not take much imagination to construct hypothetical examples in which one distribution holds more cases than another in *any* combination of adjacent categories (thus displaying more agreement) while at the same time its standard deviation is larger. The obvious explanation for such situations is the conjunction of on the one hand a fixed number of categories and on the other hand differences in the location of the central tendency of the distribution. When the mean of a distribution is located near one of the end- poles of a rating scale, a few cases at the other end-pole contribute strongly to the standard deviation (because of their large difference

*Table II.* Cumulative frequency distributions of mode plus most populous adjacent categories (using the data from Table 1)

| PvdA | | D66 | |
|---|---|---|---|
| Categories | Cum. % | Categories | Cum. % |
| Mode (7) | 41.6% | Mode (5) | 29.3% |
| Previous + (6) | 72.0% | Previous + (6) | 56.3% |
| Previous + (5) | 85.5% | Previous + (4) | 77.3% |
| Previous + (4) | 91.7% | Previous + (7) | 87.6% |
| Previous + (3) | 94.9% | Previous + (3) | 95.8% |
| Previous + (2) | 97.7% | Previous + (2) | 98.1% |
| Previous + (1) | 100.0% | Previous + (1) | 100.0% |

from the mean), while in a less skewed distribution the maximum difference from the mean is necessarily smaller, owing to which a less peaked distribution can still yield a smaller standard deviation.[3] The conclusion from this is simple: in the case of rating scales with a fixed number of categories the standard deviation not only reflects dispersion but skewedness as well, and is therefore inappropriate as measure of dispersion (or its inverse, i.e., agreement or consensus). Obviously, this conclusion not only pertains to the standard deviation, but also to measures that are transformations of it, such as Granberg and Holmberg's coefficient of consensus.[4]

A search of the literature did not reveal the existence of any satisfactory alternative measures of agreement or consensus, that are not based on the standard deviation of the observed distribution. In the following sections I propose such an alternative, the coefficient of agreement A. In addition to remedying the defects of standard deviation based measures, it has a number of advantages, such as fixed upper and lower bounds (+1 and −1, respectively), intuitively appealing interpretability of the values +1, 0 and −1, comparability across rating scales of different lengths, and an algorithm that can be implemented in a simple fashion by means of a spreadsheet or visual basic program.[5]

## 2. Agreement in Ideal-Type Distributions

In this section a measure of agreement, to be referred to as $A$, will be developed for (ordered) rating scales. The strategy employed consists of decomposing empirical distributions into constituent parts that are of an ideal-type form for which the degree of agreement can be defined in an intuitive and unambiguous way. The degree of agreement of the entire distribution can then be defined as the average agreement of its constituent parts, weighted according to the proportion of cases

they represent. For reasons of exposition I deal first with unimodal distributions, after which I extend the argument to multimodal ones.

I start by defining agreement in ideal-type distributions. One ideal-type distribution, characterized by all observations being situated in a single category (irrespective of which category this is), reflects perfect agreement. The agreement in such a distribution will be defined as 1 ($A = 1$). A contrasting ideal-type is a uniform (or rectangular) distribution, characterized by an equal number of cases in all categories of a rating scale. The degree of agreement in this case will be defined as 0 ($A = 0$). A third kind of ideal-type distribution is one that is located somewhere in between these two extremes, and is characterized by each of its categories either being empty (i.e., containing no cases at all) or alternatively containing exactly the same number of cases as each of the other non-empty categories. I will refer to such distributions as *semi-uniform*. As unimodal distributions are considered first, I consider at this stage only semi-uniform distributions of which the non-empty categories are contiguous. The degree of agreement in such distributions depends, of course, on the number of empty categories in relation to the number of non-empty ones. The smaller the number of (contiguous) non-empty categories, the larger the agreement. More explicitly, this can be expressed as follows:

$$A = 1 - \frac{(S-1)}{(K-1)}, \tag{1}$$

where $S$ is the number of non-empty categories, and $K$ is the total number of categories in the rating scale.

By evaluating this expression for $S = 1$ and for $S = K$ it can easily be seen that the ideal-type distributions of perfect agreement (all cases in 1 category) and of uniformity (each category containing the same number of cases) indeed yield respective values of 1 and 0. Furthermore, the magnitude of $A$ is not affected by the specific location of the set of (contiguous) non-empty categories. Symbolizing a semi-uniform distribution by a string of 0's (representing the empty categories) and 1's (representing the non-empty ones), the following semi-uniform distributions all yield the same value for the coefficient of agreement ($S$ = 0.667): 0000111, 0001110, 0011100, 0111000, 1110000. This is desirable as agreement should depend as little as possible on the central tendency of a distribution. A final remark on the above expression is that it implies interval assumptions (or, stated differently, an assumption of equi-distance between the categories). Obviously, whenever such assumptions are unwarranted, standard deviations are just as inappropriate as the $A$-coefficient.[6]

## 3. Decomposing Empirical Distributions into Semi-Uniform Components

Every empirical distribution of cases over the categories of a rating scale can be decomposed into a set of semi-uniform distributions, which, for reasons flowing from the decomposition procedure to be described below, may also be referred to

as *layers*. Each layer can be characterized by its specific sequence of empty and non-empty cells. Because in each layer the number of cases is identical in all non-empty cells, a layer can be represented by a *pattern* of zero's (indicating the empty cells) and 1's (indicating the non-empty ones) and by a *weight* that represents the total number of cases contained in the non-empty cells. When requiring the decomposition to result in the smallest possible number of layers, the procedure described below yields a unique and unambiguous decomposition.

The desired decomposition of an empirical distribution can be achieved in a bottom-up fashion. The first layer contains in each of the categories either zero cases (if the original empirical distribution contains zero cases in that category) or a frequency which is equal to that of the smallest non-zero category in the original distribution. This first layer is subtracted from the original distribution, and the remainder is further decomposed in the same way, until the remainder itself constitutes a layer in the sense defined above. An example of such decomposition of a hypothetical distribution in a 7- point rating scale is presented in Table III. For the purpose of illustration, the distribution is a multimodal one (it is not single-peaked), although the definition of such distributions is only discussed in the next section.

Table III illustrates how every empirical distribution in a $k$-point scale can be decomposed into at most $k$ layers. This number will be smaller when 2 or more categories in the original distribution contain exactly the same number of cases, or when the original distribution contains empty categories.

The advantage of such a decomposition is that the infinite number of different empirical distributions can be described in terms of a limited number of 'building blocks' i.e., layers characterized by their patterns. This reduces the task of expressing agreement of empirical distributions to that of expressing the degree of agreement in its constituent parts. How to do so has been defined in expression (1), above, at least for layers of which the non-empty cells are contiguous, i.e., layers deriving from unimodal distributions. I will now turn to layers in which not all non-empty categories are adjacent. Such layers are the result of decomposing multi-modal distributions according to the procedure just described.

## 4. Patterns from Multi-Modal Distributions

When an empirical distribution is not perfectly unimodal, its decomposition will necessarily generate layers in which empty categories are located in between non-empty ones. For brevity's sake such layers (and their concomitant patterns of 0's and 1's) will themselves be referred to as multi-modal. Such multi-modality may be regarded as indicative of not just lack of agreement but even of actual disagreement.

Obviously, just as the extent of agreement may vary, so may the degree of disagreement. In view of the fact that agreement in a uniform distribution has earlier been defined as 0, strong disagreement will have to be expressed by negative values of $A$.

*Table III.* Decomposition of a frequency distribution in layers, and representation by patterns and weights (*hypothetical* example, containing multi-modality)

| A | B | C | D | E | F | G | Labels for reference | A | B | C | D | E | F | G | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | Rating scale categories | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
| 30 | 40 | 210 | 130 | 530 | 50 | 10 | Observed frequencies | | | | | | | | |
| (Semi-)uniform layers in terms of frequencies | | | | | | | | Patterns (non-empty or empty cells in layer) | | | | | | | Pattern weight |
| 10 | 10 | 10 | 10 | 10 | 10 | 10 | | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0.07 |
| 20 | 20 | 20 | 20 | 20 | 20 | 0 | | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0.12 |
| 0 | 10 | 10 | 10 | 10 | 10 | 0 | | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0.05 |
| 0 | 0 | 10 | 10 | 10 | 10 | 0 | | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0.04 |
| 0 | 0 | 80 | 80 | 80 | 0 | 0 | | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0.24 |
| 0 | 0 | 80 | 0 | 80 | 0 | 0 | | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0.16 |
| 0 | 0 | 0 | 0 | 320 | 0 | 0 | | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0.32 |
| 30 | 40 | 210 | 130 | 530 | 50 | 10 | Totals | | | | | | | | 1.00 |

To elaborate the problem, let us consider the following example, that consists of patterns from a 7-point rating scale: 1100000, 1010000, 1001000, 1000100, 1000010 and 1000001. The first of these patterns reflects a (semi-)unimodal distribution (the non-empty categories are contiguous), with 5 out of 7 categories being empty. This represents a relatively high degree of agreement, $A = 0.83$. The last of these patterns expresses the most extreme expression of disagreement that this rating scale is able to register: exactly half of the cases being located in each of the extreme polar categories of the rating scale, with all intermediate categories being empty. Each of the patterns above expresses a certain mix of agreement and disagreement. The first contains no disagreement (although agreement is not perfect). The last pattern expresses maximum disagreement, and the intermediate patterns express certain degrees of both. According to the logic of this argument, agreement is stronger (and disagreement less) in (for example) the pattern 1010000 when compared to 1000100. One could also say that the deviation from unimodality is larger in the second of these two patterns. It will be apparent that the larger the number of 0's that is embedded in a pattern in between 1's, the larger this deviation from unimodality. On the other hand, (parts of) patterns in which 0's either follow or precede a series of 1's are indicative of unimodality. These notions can simply be expressed in terms of triples of categories. Only triples consisting of two 1's and one 0 can indicate deviations from unimodality, and such a deviation only exists if it displays a 101 pattern, not when it is either 110 or 011.[7] These intuitive notions can be expressed in a definition of a measure of unimodality, $U$. This measure, in turn, is to be used in conjunction with expression (1) in order to arrive at a definition of the coefficient of agreement, $A$, that covers unimodal as well as multimodal distributions and patterns. Obviously, the definition of $U$ can be specified in such a way that the resulting coefficient $A$ acquires desired properties, such as invariant and intuitively interpretable upper and lower bounds. Of these, the upper bound has already been discussed, namely that $A = 1$ in case of complete agreement. An intuitively interpretable lowerbound would be $A = -1$ in case of complete disagreement. These considerations result in the following definitions of $U$ and $A$:

For any pattern containing both 0's and 1's:

$$U = \frac{(K-2)\cdot \mathrm{TU} - (K-1)\cdot \mathrm{TDU}}{(K-2)\cdot (\mathrm{TU} + \mathrm{TDU})}, \tag{2a}$$

where TU is the number of triples of categories conforming to unimodality, and TDU is the number of triples of categories deviating from unimodality.

For any pattern containing only 1's (hence TU = TDU = 0):

$$U = 1. \tag{2b}$$

Agreement is now defined as follows:

$$A = U \cdot \left(1 - \frac{(S-1)}{(K-1)}\right), \tag{3}$$

where $S$ is the number of non-empty categories, and $K$ is the total number of categories in the rating scale.

The application of these definitions[8] and their consequences can simply be illustrated in terms of all possible patterns deriving from a 7-point rating scale (excluding mirror- images) containing exactly two 1's. The categories of the rating scale will again be referred to by the letters A to G, the patterns by strings of 1's and 0's, as in Table 3.

- **1 1 0 0 0 0 0**

This pattern contains 5 relevant triples of categories (i.e., triples consisting of two 1's and one 0): ABC, ABD, ABE, ABF, ABG. Each of these conforms to unimodality, hence TU = 5, TDU = 0, $U = 1$ and $A = 0.83$.

- **1 0 1 0 0 0 0**

Of the 5 relevant triples of categories 4 are indicative of unimodality (ACD, ACE, ACF, ACG) and 1 (ABC) deviates from unimodality. Thus: TU = 4, TDU = 1, $U = 0.56$ and $A = (0.56)(0.83) = 0.47$.

- **1 0 0 1 0 0 0**

TU = 3 (ADE, ADF, ADG), TDU = 2 (ABD, ACD). Consequently, $U = 0.12$ and $A = (0.12)(0.83) = 0.10$.

- **1 0 0 0 1 0 0**

TU = 2 (AEF, AEG), TDU = 3 (ABE, ACE, ADE), and therefore $U = -0.32$ and $A = -0.26$.

- **1 0 0 0 0 1 0**

TU = 1, TDU = 4, $U = -0.76$ and $A = -0.63$.

- **1 0 0 0 0 0 1**

TU = 0, TDU = 5, $U = -1.2$ and $A = -1$.

Using expressions (2a,b) and (3) the degree of agreement can be assessed for any pattern of 1's and 0's. The definitions and the illustration above give rise to three additional remarks:

(a) The number of relevant triples to be used in the coefficient of unimodality, $U$ (only those triples containing two 1's and one 0) is a function of the length of the scale ($K$) and the number of non-empty categories (1's) in a pattern, irrespective of their location in the rating scale. In general, for a rating scale consisting of $K$ categories, a pattern that contains $M$ 1's ($M = 2, \ldots, (K-1)$), contains $\frac{M}{2}(M-1)(K-M)$ relevant triples.

(b) How strongly a deviation from unimodality affects agreement depends not only on the number of 0's embedded in between 1's, but also on the length (i.e., the number of categories) of a rating scale. The smaller the number of categories, the stronger the effect of multimodal triples on the value of the agreement coefficient. A pattern containing only a single 101 triple results in $A = 0.47$ for a 7-point scale, but for a 5-point and 3-point rating scale it results in $A = 0.11$ and $A = -1$ respectively. This conforms to the intuitive idea that not only the absolute distance (the number of 0's between 1's) between the various modi should be reflected in $A$, but their relative distance as well.

(c) The example presented above demonstrates the desirable property that in comparable cases (that is, in all cases with identical TU + TDU) the following phenomena are perfectly linearly related: decreases in unimodality (i.e., TU), increases in multimodality (TDU), and variations in $U$ (hence also in $A$).

## 5. Summing and Weighting: The Coefficient of Agreement for Empirical Distributions

In the sections above I demonstrated that empirical distributions of observations in the categories of rating scales can always be decomposed into layers, which can – without loss of information – be represented by (ordered) patterns consisting of 0's and 1's with associated weights. Furthermore, I showed that each of such patterns can be assigned a value for agreement that conforms to a number of desirable properties, such as a close correspondence to intuitive notions of perfect (dis)agreement, insensitivity for the length of rating scales and insensitivity to the location of the empty and non-empty categories (given the degree of unimodality of a pattern and the number of its non-empty categories). As the original empirical distribution is decomposed into layers, the degree of its agreement can be described as the weighted average of the agreement in each of its constituent layers (from here on to be distinguished by an index $i$), where the proportions of cases contained in the layers are used as weights:

$$A = \sum_i w_i \cdot A_i, \tag{4}$$

where $i$ is an index for distinguishing the layers ($i = 1, \ldots, k$), $w_i$ is the proportion of cases of the empirical distribution that is contained in layer $i$, and $A_i$ is defined according to expression (3) above

## 6. Agreement for a Set of Objects

In some cases one's interest may be focussed on the agreement for a set of objects, for each of which agreement can be assessed in the manner described above. One may think of, e.g., the question to which extent voters agree in their perceptions on the ideological position of *all* parties. Such a focus shifts the level of analysis from

*Table IV.* Agreement of perceptions of left/right positions of political parties and of the Dutch party system, using proportion of valid votes obtained in the 1981 Parliamentary election as weights (parties that acquired seats indicated by *. Data: Dutch Parliamentary Election Study 1981)

| Party | Perceptual agreement (A) | Proportion of valid votes in 1981 |
|---|---|---|
| CDA* | 0.50 | 0.308 |
| PvdA* | 0.60 | 0.283 |
| VVD* | 0.55 | 0.173 |
| D66* | 0.56 | 0.111 |
| PPR* | 0.62 | 0.020 |
| CPN* | 0.85 | 0.021 |
| PSP* | 0.70 | 0.021 |
| DS70 | 0.33 | 0.006 |
| SGP* | 0.57 | 0.020 |
| BP | 0.47 | 0.002 |
| GPV* | 0.59 | 0.008 |
| RPF* | (not asked) | 0.012 |
| Other parties | (not asked) | 0.015 |
| Overall | 0.55 | |

a single object (a single political party) to a system (in this example: a political system).

The perceptual agreement of a set of objects may be defined as the appropriately weighted average of the degrees of perceptual agreement for the constituent objects of the set.[9] For the example of perceptions of political parties the first step is to determine the degree of perceptual agreement for each of the political parties (see Table IV). Subsequently, appropriate weights have to be chosen for combining these into an overall measure. Here, of course, the substantive nature of the research question addressed is of paramount importance. In some instances, the most appropriate weight will be the proportion of valid votes acquired by a party in the most recent national election (see, e.g., Van der Eijk and Franklin, 1996; Oppenhuis, 1995). In other situations, one could think of other criteria, such as the proportion of seats parties hold in parliament, the amount of attention media give to parties, the money they spend on paid publicity in election campaigns, etc. An example, based on 1981 data, is presented in Table IV.

The survey that provided the data for Table III contained a 10-category left/right scale that was used to let respondents indicate the position of 11 different parties.

Although this is a large number, it failed to include the RPF, a new party that in 1981 for the first time acquired parliamentary representation. On the other hand, two of the parties that were included (DS70 and BP) obtained too few votes for getting even a single seat in parliament. Moreover, a number of other parties participated in the 1981 election without even coming close to obtaining a single seat. Collectively they polled no more than 1.5% of the valid votes. Fortunately, the parties included in this survey question represent between them more than 97% of both votes cast and seats in parliament, so that – given the weights applied – the survey information comes quite close to a comprehensive coverage of the Dutch party system.

The possibility to assess the perceptual agreement of sets of objects is particularly relevant for comparative research. One form this may take is the longitudinal comparison of a single system, in order to, for example, assess the changing relevance of ideological or issue dimensions for cognitions and evaluations (see, e.g., Van der Brug, 1997). A different application is the comparison of the importance of such dimensions in different political systems (cf. Oppenhuis, 1995; Van der Eijk and Franklin, 1996, Van der Eijk et al., 1999). As a case in point, the European Election Study 1989 contained information on the voters' perception of the ideological (left/right) position of political parties in their political system at the time of the direct elections to the European Parliament in June 1989.[10] Table V reports the degree of perceptual agreement in each of the systems of the (then) European Community. At a merely descriptive level, Table V displays that voters agree in most member states of the European Community to a rather large extent on how they describe parties in terms of left/right. This allows the left/right imagery to function as an effective communication device in politics. At the same time, Table V also shows considerable differences between countries in terms of perceptual agreement, that is in the extent to which different voters in a country apply this yardstick in the same way.

## 7. Empirical Applications

In a 1990 paper I compared voter behavior in the member states of the European Community. I found perceptual agreement of parties' ideological (left/right) positions to be strongly related to other systemic factors such as the proportion of voters behaving according to Downs' (1957) smallest ideological distance theorem (while controlling for the number of parties in a system). In a more generalized fashion, such perceptual agreement was also found to be strongly related to the extent to which party choice is determined by ideological proximity. Finally, the adequacy of ideological representation of voters by political parties in parliament was also found to be strongly correlated with perceptual agreement.[11] The strength of the intercorrelations between these various phenomena is reported in Table VI.

Partly building on these earlier findings, a team of researchers studying party choice comparatively in the member-states of the European Union, used the

*Table V.* Agreement of perceptions of left/right positions of parties in political systems in the European Community. June 1989 (data: European Election Study 1989; see Van der Eijk et al., 1993; Van der Eijk and Franklin, 1996)

| | |
|---|---|
| Belgium (Flanders) | 0.40 |
| Belgium (Wallonia) | 0.40 |
| Denmark | 0.63 |
| France | 0.65 |
| Germany | 0.56 |
| Greece | 0.71 |
| Ireland | 0.48 |
| Italy | 0.60 |
| Luxembourg | 0.52 |
| The Netherlands | 0.54 |
| Portugal | 0.63 |
| Spain | 0.48 |
| United Kingdom (Great Britain) | 0.55 |
| United Kingdom (Northern Ireland) | 0.36 |

*Table VI.* Correlations between propensity to vote according toward ideological proximity, adequacy of left-right representation, and left-right perceptual agreement ($n = 13$). (Source: Van der Eijk, 1990.)

| | | | |
|---|---|---|---|
| Ideological proximity | 1.0 | | |
| Representation | 0.85 | 1.0 | |
| Perceptual agreement | 0.87 | 0.77 | 1.0 |
| | id.prox. | repres. | agreem. |

concept of perceptual agreement to characterize political cultures, and to explain why the causal effect of various determinants of party choice differs in strength between member states of the European Union. Both Erik Oppenhuis' *Voting Behavior in Europe* (1995) and Cees van der Eijk and Mark Franklin's *Choosing Europe*? (1996) deal with these questions. When focussing in particular on agreement of perceptions of left/right positions of political parties, these authors find that the stronger such agreement, the stronger the causal effect of left/right proximity on voters' party preferences.[12] In other words, the stronger perceptual agreement of parties' ideological positions, the stronger the determinant of party choice that may be called rational-ideological in the Downsian meaning of the

term (Downs, 1957). At the same time these authors find that the causal effect of issue preferences on party preferences diminishes with increasing perceptual agreement on the ideological position of parties. These findings help to understand why, when taken separately, causal effect parameters of ideology and issues differ in strength across political systems. Moreover they also provide empirical support for Downs' interpretation of ideologies as information-cost saving devices (Downs, 1957: 96–113).[13]

Van der Brug took perceptions of parties as the dependent variable in his *Where's the Party?* (1997). He asks to which extent voter perceptions of parties' positions on issues and in ideological terms are adequate and shared between voters. This interest flows from a concern with democratic representation. The more adequate perceptions are and the more people agree on how they perceive the political world, the more elections can be expected to function as effective channels of communication and control between citizens and elites (see also Converse, 1964, 1975). In this over-time comparative study on the Netherlands, Van der Brug finds that the degree of perceptual consensus is clearly related to the prominence of an issue on the political agenda. This relation apparently is the consequence of the contents of information flows and of political communication processes. A second important determinant of perceptual agreement is found to be the equivocality of parties' reactions to political events.

Although the previous paragraphs can in no way do justice to the richness of the publications referred to, they illustrate clearly that perceptual agreement is strongly linked to processes of political communication and information, to the dynamics of public opinion, to the actual behavior of individual voters, and to the electoral processes at large. Moreover, they illustrate that in order to observe these effects, an adequate measure of agreement is required, and that the coefficient defined in this paper is more adequate than other ones commonly found in the literature.

## Acknowledgements

The coefficient described in this paper has been developed during a stay at NIAS – the Netherlands' Institute for Advanced Study in the Social Sciences and Humanities – in 1988–1989. It has since then been used extensively and productively in many research projects (cf. Van der Eijk, 1990; Oppenhuis, 1995; Van der Eijk and Franklin, 1996; Van der Brug, 1997), but has not been properly described in a form accessible to those not part of my network of direct research contacts.

In addition to my gratitude to NIAS for offering a climate for productive research, I want to express my indebtedness to Mark Franklin, Erik Oppenhuis and Wouter van der Brug for their comments on earlier versions of this manuscript and for their contributions to the substantive applications reported in the final sections of this paper.

## Notes

1. Such rating scales are often somewhat problematic because of extreme clustering of responses in a small number of disconnected categories (usually in the multiples of 10, and in categories 25 and 75). Good reasons exist for not accepting such findings as evidence of true multimodality in the distribution. Therefore, one often finds that the responses to such scales are recoded in a smaller number of categories. Such recoded scales lend themselves much better for calculating agreement by the coefficient proposed here.
2. Thomassen and Jennings (1989) reject standard deviations and the Granberg–Holmberg coefficient for the same reasons as put forward in this paper. They propose as an alternative a measure of homogeneity that consists of the proportion of cases in two categories: the modal one and the category adjacent to the mode that contains the largest number of cases. Obviously, this measure has two defects. First, it becomes less informative the lower it is, as it is unable to reflect how the remainder of the cases are distributed in relation to those in the two categories involved: unimodal, multimodal, etc. Second, the magnitude of this measure can be expected to be dependent on the length of the rating scale (the number of categories), which makes it less suited for comparisons across different scales. The measure proposed in this paper does not suffer from these deficiencies.
3. This phenomenon has more often been noticed in the context of summarizing the central tendency of a distribution of observations in an ordered rating scale with a limited number of categories. From that, the –justified – conclusion has been drawn that the arithmetic mean is a less appropriate measure of the central tendency of the kind of rating scales discussed here than the interpolated median is, cf. Huber and Bingham-Powell (1994) and Herrera et al. (1992).
4. Granberg and Holmberg's coefficient of consensus is defined as $1 - \frac{S_o}{S_u}$, where $S_o$ indicates the observed standard deviation of a distribution in a rating scale of fixed length, and $S_u$ indicates the standard deviation of a uniform distribution in the same rating scale. In the remainder of this paper I will refer to this coefficient as the G–H coefficient.
5. A dedicated spreadsheet for rating scales of 3, 4, 5, 7 or 10 categories can, upon request, be obtained from the author. Analogous applications for rating scales of other lengths can simply be made by adapting any of these.
6. The strategy used in this paper to define a coefficient of agreement allows in principle that knowledge about unequal distances between the categories of a rating scale is taken into account. At this place I will not elaborate upon such possibilities.
7. When looking at deviations from unimodality, only those triples of categories have to be taken into account that consist of two 1's and one 0. All other kinds of triples are by definition unable to indicate deviations from unimodality. Three 0's, three 1's or one 1 with two 0's can never indicate a breach of unimodality, which is easily confirmed by visual inspection of such patterns: 000, 111, 100, 010, 001. The reader who is familiar with parallelogram and unfolding analysis will have noticed the similarity between the approach utilized here and the way in which those methods define 'error' in individual response patterns (refer to, amongst others, Coombs, 1964; Van Schuur, 1984, 1993).
8. In an earlier version of this manuscript and the algorithm based on it, $U$ was defined somewhat differently than in this text, namely

$$U = \frac{\text{TU}-\text{TDU}}{\text{TU}+\text{TDU}}.$$

The definition of $A$ was identical, according to expression (3). This previous definition ensured that $U$ itself had a lowerbound of $-1$ and an upperbound of 1, but its main disadvantage was the lack of a fixed lowerbound for $A$, which is, incidentally, equally true for the G–H coefficient discussed earlier. I am grateful to Wouter van der Brug for pointing this out and

for his contribution to a remedy for this. A perfectly polarized distribution (all cases equally split between only the two extreme categories of a rating scale) yielded, with this previous definition the following values for respectively 4-point, 5-point, 7-point and 10-point scales: $-0.67$, $-0.75$, $-0.83$ and $-0.89$. The corresponding values of the G–H coefficient are $-0.35$, $-0.42$, $-0.5$ and $-0.57$. The definitions presented in the text ensure that in all these cases $A = -1$, a commonality by which comparison of agreement of strongly polarized distributions between rating scales of different lengths is greatly facilitated.

9. Obviously, for a given group of people, we can also consider their preferential (rather than perceptual) agreement for a set of objects. In that case, the aggregation would involve different stimuli, each represented by its own rating scale on which people have to indicate their own most preferred position. The elaboration of such a possibility runs entirely parallel with the discussion in the text on perceptual agreement, and will therefore not be discussed separately.
10. See for details about this study: Van der Eijk et al. (1993).
11. 'Adequacy of ideological representation' refers to the availability of parties at close ideological (i.e., left/right) distance from the voter, irrespective of whether (s)he votes for that party.
12. The same result has been found in a replicatory analysis on the basis of the 1994 European Election Study, cf. Van der Eijk et al. (1999). For the documentation of this study, see: Schmitt et al. (1996).
13. These results could not be obtained when using a coefficient of 'consensus' based on standard deviations of the empirical distribution, such as the G–H coefficient discussed in Section 1 of this article. In view of the interconnection and compatibility of the findings reported in the text with well-established other findings in the area of party choice research, I consider the findings obtained with the coefficient of agreement A as valid, and the inability to find these results when using a different measure of agreement as evidence of the latter's defects, particularly in combination with their defects discussed in Section 1 of this article.

## References

Brug, W. van der (1997). *Where's the Party? Voters' Perceptions of Party Positions*. Doctoral Dissertation: University of Amsterdam.

Brug, W. van der & van der Eijk, C. (1999). The cognitive basis of voting. In: H. Schmitt & J. J. A. Thomassen (eds), *Political Representation and Legitimacy in the European Union*. Oxford: Oxford University Press, pp. 129–160.

Converse, P. E. (1964). The nature of belief systems in mass publics. In: D. E. Apter (ed.), *Ideology and Discontent*. New York: Free Press, pp. 206–261.

Converse, P. E. (1975). Public opinion and voting behavior. In: F. I. Greenstein & N. W. Polsby (eds), *Handbook of Political Science*, Vol. 4, Reading, MA: Addison Wesley, pp. 75–169.

Converse, P. E. & Pierce, R. (1986). *Political Representation in France*. Cambridge, MA: Belknap Press.

Coombs, C. H. (1964). *A Theory of Data*. New York: Wiley.

Downs, A. (1957). *An Economic Theory of Democracy*. New York: Harper and Row.

Eijk, C. van der (1990). Ideology, party systems and voting in Europe. Paper presented at Annual Meeting of the American Political Science Association, San Francisco, August 1990.

Eijk, C. van der & Franklin, M. N. (1996). *Choosing Europe? The European Electorate and National Politics in the Face of Union.* Ann Arbor: University of Michigan Press.

Eijk, C. van der, Franklin, M. N. & van der Brug, W. (1999). Policy preferences and party choice. In: H. Schmitt & J. J. A. Thomassen (eds), *Political Representation and Legitimacy in the European Union*. Oxford: Oxford University Press, pp. 161–185.

Eijk, C. van der, Niemöller, B. & Eggen, A. Th. J. (1981). *Dutch Parliamentary Election Study 1981*. Amsterdam: University of Amsterdam.

Eijk, C. van der, Oppenhuis, E. V. & Schmitt, H. (1993). *EES-89: European Election Study 1989. Data Description and Preliminary Documentation*. Amsterdam: Steinmetz Archive/Swidoc.

Granberg, D. & Holmberg, S. (1988). *The Political System Matters. Social Psychology and Voting Behavior in Sweden and the United States*. Cambridge: Cambridge University Press.

Herrera, C., Herrera, R. & Smith, E. (1992). Public opinion and congressional representation. *Public Opinion Quarterly* 56: 185–205.

Huber, J. & Bingham Powell, G. (1994). Congruence between citizens and policymakers in two visions of liberal democracy. *World Politics* 46: 291–326.

McClosky, H. (1964). Consensus and ideology in American politics. *American Political Science Review* 58: 361–382.

McClosky, H., Hoffman, P. J. & O'Hara, R. (1960). Issue conflict and consensus among party leaders and followers. *American Political Science Review* 54: 406–427.

Schmitt, H., van der Eijk, C., Scholz, E. & Klein, M. (1996). *European Election Study 1994. Data Description and Documentation*. Mannheim/Cologne, Zentralarchiv für Empirische Sozialforschung/European Election Study Research Group.

Schuur, W. H. van (1984). *Structure in Political Beliefs. A New Model for Stochastic Unfolding with Applications to European Party Activists*. Amsterdam: CT Press.

Schuur, W. H. van (1993). Nonparametric unfolding models for multicategory data. *Political Analysis* 4: 41–74.

Thomassen, J. J. A. & Jennings, M. K. (1989). *Party Systems, Party Differences and Homogeneity*. Paper presented at IPSA/ISA Workgroup on Elections and Parties, Paris, April 1989.

Thomassen, J. J. A. & Schmitt, H. (1999). Partisan structures in the European parliament. In: R. S. Katz & B. Wessels (eds), *The European Parliament, the National Parliaments and European Integration*. Oxford: Oxford University Press, pp. 129–148.

Oppenhuis, E. V. (1995). *Voting Behavior in Europe. A Comparative Analysis of Electoral Participation and Party Choice*. Amsterdam: Het Spinhuis.

Pomper, G. M. (1972). From confusion to clarity: Issues and American voters, 1956–1968. *American Political Science Review* 66: 415–428.